
BLUE STOCK

Mutual Fund Analytics

Final Project Report

Field	Details
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Period	Summer 2025
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Project	Bluestock Mutual Fund Analytics

Executive Summary

This report presents the complete findings of the Bluestock Mutual Fund Analytics project — a comprehensive, end-to-end data analytics engagement covering 40 mutual fund schemes across 10 Asset Management Companies (AMCs) in India. The project was completed as part of the Bluestock Fintech Summer Internship 2025 at BML Munjal University.

The project processed 46,000 NAV records, 32,778 investor transactions, and quarterly AUM data spanning January 2022 to May 2026. It delivered a complete analytics platform including an ETL pipeline, a SQLite star-schema database, exploratory data analysis with 15+ charts, quantitative performance metrics, an interactive Power BI dashboard, and advanced risk analytics including Value at Risk (VaR), investor cohort analysis, and a fund recommender system.

Key findings reveal that industry SIP inflows grew 3x to ₹31,002 Crore, industry folios doubled to 26.12 Crore, SBI Mutual Fund dominates AUM at ₹12.5 Lakh Crore, and the average fund alpha of +1.254% confirms genuine outperformance beyond market returns during the analysis period.

1. Introduction

1.1 Background

The Indian mutual fund industry has witnessed unprecedented growth over the past four years, with total industry AUM crossing ₹50 Lakh Crore and monthly SIP inflows reaching record highs. This project aims to build a comprehensive analytics platform for Bluestock Fintech that can serve as a foundation for investor advisory, fund comparison, and risk management.

1.2 Objectives

- Build an automated ETL pipeline to ingest, clean, and validate mutual fund data
- Design and implement a relational SQLite database with a proper star schema
- Perform exploratory data analysis to uncover industry trends and investor patterns
- Compute quantitative performance metrics including Sharpe, Sortino, Alpha, Beta, and VaR
- Build an interactive Power BI dashboard for business stakeholders
- Develop advanced analytics including a fund recommender and investor cohort analysis

1.3 Scope

The project covers 40 mutual fund schemes across 10 AMCs, 12 sub-categories, and 5 risk grades. Data spans January 2022 to May 2026, covering equity, debt, liquid, and index fund categories in both Regular and Direct plans.

2. Data Sources & Architecture

2.1 Datasets

Ten structured CSV datasets were provided as the primary data source, supplemented by live NAV data fetched from the mfapi.in public API for 6 key schemes:

#	Dataset	Records	Key Columns
01	01_fund_master.csv	40 rows	amfi_code, fund_house, category, risk_grade
02	02_nav_history.csv	46,000 rows	amfi_code, date, nav
03	03_aum_by_fund_house.csv	90 rows	date, fund_house, aum_crore
04	04_monthly_sip_inflows.csv	48 rows	month, sip_inflow_crore, yoy_growth_pct
05	05_category_inflows.csv	144 rows	month, category, net_inflow_crore
06	06_industry_folio_count.csv	21 rows	month, total_folios_crore
07	07_scheme_performance.csv	40 rows	amfi_code, sharpe_ratio, alpha, beta
08	08_investor_transactions.csv	32,778 rows	investor_id, transaction_type, amount_inr
09	09_portfolio_holdings.csv	322 rows	amfi_code, sector, weight_pct
10	10_benchmark_indices.csv	8,050 rows	date, index_name, close_value

2.2 Data Quality

The raw datasets were of high quality with minimal issues. Key anomalies detected and resolved:

- 04_monthly_sip_inflows: 12 NULL values in yoy_growth_pct for Jan–Dec 2022 (expected — no prior year baseline available)
- All date columns stored as strings — converted to datetime using pd.to_datetime() during cleaning
- 02_nav_history: Forward-filled for weekends and public holidays using ffill() after reindexing to full date range — added 18,320 gap rows
- All other datasets: Zero nulls, zero duplicates across 10 datasets

2.3 SQLite Star Schema

A relational star schema was designed and implemented in bluestock_mf.db with the following structure:

- Dimension tables: dim_fund (40 schemes), dim_date (calendar dates with year/month/quarter/weekday attributes)
- Fact tables: fact_nav (daily NAV), fact_transactions (investor activity), fact_performance (risk metrics), fact_aum (quarterly AUM), fact_sip_inflows, fact_category_inflows, fact_portfolio_holdings, fact_benchmark_indices
- All fact tables linked to dim_fund via amfi_code and to dim_date via date_id as foreign keys
- 10 analytical SQL queries written covering AUM ranking, NAV trends, SIP growth, sector analysis, and risk-adjusted performance

3. Exploratory Data Analysis

3.1 Industry Overview

The Indian mutual fund industry showed remarkable growth across all metrics during the analysis period of 2022 to 2026:

Metric	Value	Trend
Monthly SIP Inflow (Dec 2025)	₹31,002 Crore	3x growth from Jan 2022
Total Industry Folios (Dec 2025)	26.12 Crore	Doubled from 13.26 Crore
Largest AMC by AUM	SBI Mutual Fund	₹12.5 Lakh Crore (2025)
Total Schemes Analysed	40 schemes	10 AMCs, 12 sub-categories
Transaction Volume	32,778 records	SIP: 60.2%, Lumpsum: 24.7%
Data Coverage	Jan 2022 – May 2026	4.4 years, 46,000 NAV records

3.2 NAV Trends

The daily NAV trend analysis revealed two distinct market phases. The 2023 bull run saw consistent NAV appreciation across all Large Cap Direct schemes, with most funds more than doubling their January 2022 NAV values by December 2023. The 2024 market correction (September to November 2024) caused visible NAV drawdowns of 8–15% across equity funds before recovery resumed in 2025.

3.3 AUM Growth

SBI Mutual Fund leads all AMCs with ₹12.5 Lakh Crore AUM as of 2025, nearly twice HDFC Mutual Fund at approximately ₹6.8 Lakh Crore. All 10 AMCs showed consistent year-on-year AUM growth from 2022 to 2025, reflecting strong net inflows and market appreciation.

3.4 Investor Demographics

Analysis of 32,778 investor transactions revealed clear demographic patterns:

- Age: 26–35 age group dominates SIP participation (approximately 38% of all SIP investors)
- Gender: Male investors at approximately 62% vs Female at 38%
- Geography: Maharashtra leads all states in total SIP amount; T30 cities contribute 78% of total SIP volume vs B30 at 22%
- Payment: UPI is the dominant payment mode for SIP transactions
- KYC: 92% of investors have Verified KYC status (30,146 out of 32,778 transactions)

4. Performance Analytics

4.1 Methodology

All performance metrics were computed from scratch using daily NAV data. The risk-free rate used throughout is 6.5% per annum, proxying the RBI repo rate. All metrics are annualised using 252 trading days.

Metric	Formula	Interpretation
CAGR	$(NAV_end/NAV_start)^{(1/n)} - 1$	Compounded annual growth rate for 1yr, 3yr, 5yr
Sharpe Ratio	$(R_p - R_f) / \sigma \times \sqrt{252}$	Return per unit of total risk
Sortino Ratio	$(R_p - R_f) / \sigma_{downside} \times \sqrt{252}$	Return per unit of downside risk only
Alpha	OLS intercept $\times 252$	Annualised excess return vs NIFTY100
Beta	OLS slope vs NIFTY100	Market sensitivity coefficient
Max Drawdown	$\min(NAV/running_max - 1)$	Worst peak-to-trough decline
VaR (95%)	5th percentile of daily returns	Worst daily loss at 95% confidence
CVaR (95%)	Mean of returns below VaR	Expected loss on worst days

4.2 Key Performance Findings

- Maximum 3-year CAGR: 23.39% (SBI Small Cap Fund Regular Plan Growth)
- Best Sharpe Ratio: ICICI Pru Liquid Fund Regular Growth — highest risk-adjusted returns
- Average Sharpe across all 40 funds: 1.362 — indicating generally good risk-adjusted performance
- Average Alpha: +1.254% annualised — most funds generated positive alpha vs NIFTY100
- Average Beta: 0.873 — portfolio behaves slightly defensively relative to the market
- Average Max Drawdown: -19.20% — typical equity fund lost ~19% from peak to trough

4.3 Fund Scorecard

A composite 0–100 scoring system was built combining: $30\% \times 3\text{-year CAGR rank} + 25\% \times \text{Sharpe ratio rank} + 20\% \times \text{Alpha rank} + 15\% \times \text{Expense ratio rank (inverse, lower is better)} + 10\% \times \text{Maximum drawdown rank (inverse, smaller drawdown is better)}$. Each component was converted to a percentile rank before weighting.

This scorecard provides a single defensible ranking that accounts for returns, risk efficiency, cost, and downside protection simultaneously — more meaningful than ranking by any single metric.

5. Advanced Analytics

5.1 Value at Risk and CVaR

Historical VaR at 95% confidence was computed for all 40 schemes using the 5th percentile of the empirical daily return distribution. CVaR (Conditional VaR, also known as Expected Shortfall) represents the average loss on the worst 5% of trading days.

- Small Cap and Mid Cap funds showed the highest tail risk with VaR exceeding -2.5% daily
- Liquid and Gilt funds showed minimal tail risk with VaR below -0.3% daily
- The VaR hierarchy perfectly matches the SEBI risk grade labels, validating the data quality

5.2 Rolling Sharpe Analysis

Rolling 90-day Sharpe ratios were computed for 5 key funds (Large Cap, Mid Cap, Small Cap, Flexi Cap, ELSS). The analysis confirmed the 2023 bull run produced exceptional risk-adjusted returns with Sharpe ratios crossing 2.0 for Large Cap Direct funds. The 2024 correction pulled Sharpe ratios below 1.0 for Mid and Small Cap funds while Large Cap maintained greater resilience above 0.5.

5.3 Investor Cohort Analysis

Investors were grouped by their first transaction year to understand behavioural differences across cohorts. The 2024 cohort showed the highest average SIP amount, suggesting newer investors enter with greater financial awareness and capacity. The 2022 cohort predominantly prefers Large Cap Direct funds, while newer cohorts show increasing preference for Mid Cap and Flexi Cap, indicating growing risk appetite among recent market entrants.

5.4 SIP Continuity Analysis

For investors with 6 or more SIP transactions, the average gap between consecutive transaction dates was computed. Investors with an average gap exceeding 35 days were flagged as "At Risk" — likely to pause or discontinue their SIP. The majority of regular SIP investors maintain a healthy monthly cadence, but the at-risk segment represents a meaningful retention opportunity for AMCs.

5.5 Fund Recommender

A rule-based fund recommender (recommender.py) was built that accepts risk appetite (Low, Moderate, or High) as input and returns the top 3 funds by Sharpe ratio within the matching risk grade. Low risk maps to Gilt and Liquid funds, Moderate to Large Cap and Flexi Cap, and High to Mid Cap and Small Cap. The tool is deployable as a command-line application.

5.6 Sector HHI Concentration

The Herfindahl-Hirschman Index ($HHI = \sum \text{weight}_i^2$) was computed for all equity funds at the sector level. A high HHI indicates concentrated sector bets while a low HHI indicates well-diversified holdings. Funds with high Banking and IT concentration showed elevated HHI scores, making them susceptible to sector-specific shocks.

6. Power BI Dashboard

6.1 Design Overview

A 4-page interactive Power BI dashboard (bluestock_mf_dashboard.pbix) was built connecting all 10 cleaned datasets. The data model uses dim_fund as the central dimension table linked to all fact tables via amfi_code. Industry-aggregate tables (AUM, SIP, Category, Benchmark) operate as standalone tables on their respective pages.

6.2 Dashboard Pages

Page	Title	Key Visuals	Interactivity
1	Industry Overview	KPI cards (AUM, SIP, Folios, Schemes), AUM trend line, AUM by AMC bar	Date slicer
2	Fund Performance	Risk-return scatter, Fund scorecard table, NAV vs benchmark line	Fund house, Category, Plan slicers
3	Investor Analytics	Amount by state bar, Transaction type donut, Age group bar, Monthly volume line	State, Age group, City tier slicers
4	SIP & Market Trends	SIP inflow bar, NIFTY50 line, Category heatmap matrix, Top 5 table	Category filter

6.3 Key Dashboard Insights

- Page 1 confirmed SBI Mutual Fund's AUM dominance and the consistent upward AUM trend across all AMCs
- Page 2 scatter plot clearly separates Small/Mid Cap funds (high return, high risk) from Liquid/Gilt (low return, low risk)
- Page 3 revealed Maharashtra as the leading SIP state and 26–35 as the dominant age group
- Page 4 heatmap showed Sectoral/Thematic category receiving the highest net inflows in recent months

7. Conclusions & Recommendations

7.1 Key Conclusions

- India's mutual fund industry is in a structural growth phase — SIP inflows, folio counts, and AUM all showing strong multi-year uptrends
- The positive average alpha of +1.254% across funds suggests that active management has added genuine value beyond what index returns explain, particularly in the 2022–2025 period
- Direct plans consistently outperform Regular plans on a risk-adjusted basis due to their lower expense ratios — investors in Regular plans pay 1%+ more annually in distributor commissions
- SIP is the dominant investment mode at 60.2% of transactions, reflecting healthy disciplined investing behaviour and suggesting AMCs should continue focusing on SIP retention
- Geographic concentration in T30 cities (78% of SIP volume) represents both a current strength and a significant growth opportunity in B30 markets

7.2 Recommendations

- AMCs should build targeted SIP retention campaigns for the at-risk investor segment (avg gap > 35 days) using personalized reminders and incentives
- B30 city expansion should be a strategic priority — digital platforms and simplified KYC have removed most barriers to entry for smaller city investors
- Investors with Low risk appetite should be directed toward Direct plan Gilt and Liquid funds with expense ratios below 0.5% and positive Sharpe ratios
- The Banking sector concentration (28% allocation) across equity funds creates systemic risk — investors should diversify across funds with varying sector HHI profiles
- The 26–35 age cohort is the most addressable segment for new SIP acquisition — products tailored to their digital preferences and income growth trajectory should be prioritised

7.3 Project Limitations

- NAV data covers only 40 out of 1,908+ AMFI-registered schemes — findings may not fully represent the industry
- Investor transaction data is simulated for privacy purposes — real behavioral patterns may differ
- Alpha and Beta computed against NIFTY100 only — some funds use different benchmarks (Nifty Midcap, BSE Smallcap)

8. Technical Stack

Category	Tools & Libraries	Version / Notes
Language	Python	3.10
Data Processing	Pandas, NumPy	3.0.2, 2.4.4
Visualisation	Matplotlib, Seaborn, Plotly	3.10.8, 0.13.2, 6.8.0
Database	SQLite + SQLAlchemy	2.0.51
Statistics	SciPy (linregress, percentile)	1.17.1
Notebooks	Jupyter Lab	1.1.1
Dashboard	Microsoft Power BI Desktop	June 2025
Version Control	Git + GitHub	github.com/Jatinyadav22
API	mfapi.in	Live NAV fetch for 6 schemes
HTTP	Requests	2.33.1

Section 9: Bonus Implementations

Four bonus challenges were completed beyond the core project deliverables, adding production-grade automation, web deployment, quantitative risk modelling, and portfolio optimisation capabilities.

9.1 B1 — Scheduled ETL Pipeline (Auto NAV Fetch)

File: etl_scheduler.py

A fully automated ETL scheduler was built as a single Python file that fetches live NAV data from mfapi.in every weekday at 8:00 PM without any manual intervention.

Technical Implementation:

- Uses the schedule library to trigger jobs on Monday through Friday at 20:00
- Fetches NAV for 6 key schemes via GET https://api.mfapi.in/mf/{amfi_code}
- Implements duplicate prevention — only inserts new (amfi_code + date) pairs not already in the SQLite database
- Logs every run to an etl_run_log table in the database with status, row count, and error messages
- Saves raw CSVs to data/raw/ and a summary to live_nav_summary.csv
- Automatically skips weekends and public holidays
- Registers as a Windows Task Scheduler job via --setup-windows flag so it runs even when VS Code is closed

Commands:

```
python etl_scheduler.py          # Start scheduler
python etl_scheduler.py --run-now # Test immediately
python etl_scheduler.py --status  # View last run
python etl_scheduler.py --history # View last 10 runs
python etl_scheduler.py --setup-windows # Register Windows Task
```

9.2 B2 — Streamlit Web Application

File: app.py

A fully interactive web application was built using Streamlit as a browser-based alternative to the Power BI dashboard. The app mirrors all 4 Power BI dashboard pages with real-time filtering.

Pages:

- **Industry Overview** — KPI cards (Total AUM, SIP inflow, Folios, Schemes), AUM trend, AUM by fund house, folio growth, sub-category pie

- **Fund Performance** — Interactive risk vs return scatter plot, NAV vs NIFTY100 indexed chart, Sharpe ratio ranking, full fund scorecard table, sector allocation donut
- **Investor Analytics** — Transaction amount by state, SIP/Lumpsum/Redemption donut, age group bar chart, monthly volume line, T30 vs B30 pie, payment mode chart
- **SIP & Market Trends** — SIP inflow trend with ATH annotation, benchmark index selector, SIP vs active accounts dual axis, category heatmap, top 5 categories, built-in fund recommender

Run:

pip install streamlit

streamlit run app.py

Opens at <http://localhost:8501> in your browser.

9.3 B3 — Monte Carlo NAV Simulation

File: monte_carlo.py

Output: monte_carlo_results.csv, reports/chart_monte_carlo_top5.png, reports/chart_monte_carlo_all_funds.png

A Monte Carlo simulation was built using Geometric Brownian Motion (GBM) to project NAV growth over a 5-year horizon for all 40 schemes with uncertainty bands.

Mathematical Model:

$$S(t+1) = S(t) \times \exp((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t} \times Z)$$

Where:

μ = mean daily log return (from historical data)

σ = standard deviation of daily log returns

$Z \sim N(0,1)$ independent random shock each day

$\Delta t = 1$ (daily steps)

Parameters:

- 1,000 simulation paths per fund
- 5-year horizon (1,260 trading days)
- Confidence bands at 5th, 25th, 50th, 75th, 95th percentiles
- Random seed 42 for reproducibility

Key Results (Top 5 Funds by Expected Return):

Fund	Current NAV	P50 (5yr)	P95 (5yr)	P(Profit)
ICICI Pru Midcap	₹342	₹1,676	₹2,890	99.8%
Mirae Asset Large Cap	₹332	₹813	₹1,420	100%

Fund	Current NAV	P50 (5yr)	P95 (5yr)	P(Profit)
Mirae Asset Tax Saver	₹137	₹341	₹598	99.9%
Kotak Flexicap	₹205	₹557	₹960	100%
SBI Small Cap	₹442	₹1,026	₹1,890	98.2%

Insight: All equity funds show greater than 98% probability of profit over a 5-year horizon, reinforcing the case for long-term SIP investing over short-term trading.

9.4 B4 — Markowitz Efficient Frontier Optimisation

File: markowitz.py

Output: markowitz_results.csv, reports/chart_efficient_frontier.png,

reports/chart_portfolio_weights.png, reports/chart_correlation_heatmap_markowitz.png

Portfolio optimisation was implemented using the Markowitz Mean-Variance framework for 5 selected funds across different equity categories — Large Cap, Mid Cap, Small Cap, Flexi Cap, and ELSS.

Selected Funds:

Fund	Category	Ann Return	Ann Volatility
SBI Bluechip Regular	Large Cap	23.10%	13.74%
HDFC Mid-Cap Opportunities	Mid Cap	27.21%	18.94%
SBI Small Cap Regular	Small Cap	30.27%	25.14%
Kotak Flexicap Regular	Flexi Cap	27.26%	15.89%
Mirae Asset Tax Saver	ELSS	28.33%	17.67%

Methodology:

- 10,000 random portfolios generated using Dirichlet distribution (weights sum to 1)
- Efficient frontier computed using scipy.optimize.minimize with SLSQP method
- Constraints: weights sum to 1, each fund capped at 40% (no short selling)
- Capital Market Line computed from risk-free rate (6.5%) to Max Sharpe portfolio

Optimisation Results:

Portfolio	Annual Return	Volatility	Sharpe Ratio
Maximum Sharpe	26.70%	7.57%	2.668
Minimum Variance	26.42%	7.52%	2.649
Equal Weight (20% each)	27.23%	8.25%	2.514

Optimal Weights (Maximum Sharpe Portfolio):

- SBI Bluechip (Large Cap): 26.4%

-
- Kotak Flexicap: 25.8%
 - Mirae Asset Tax Saver (ELSS): 20.9%
 - HDFC Mid-Cap: 16.2%
 - SBI Small Cap: 10.7%

Key Insight: The optimised Max Sharpe portfolio achieves a Sharpe ratio of 2.668 versus 2.514 for equal weight — a 6% improvement in risk-adjusted returns simply by reallocating weights without changing the fund selection. Near-zero pairwise correlations across all 5 funds confirm strong diversification benefit.